

Process Creation

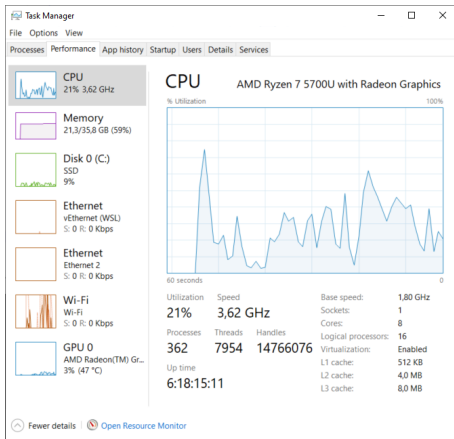
2025 Semester 1 SOFTENG 370: Operating Systems

Talia Xu

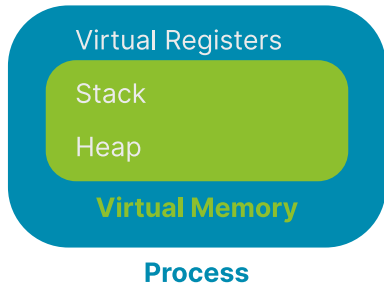
Lecture 4

2.0.0

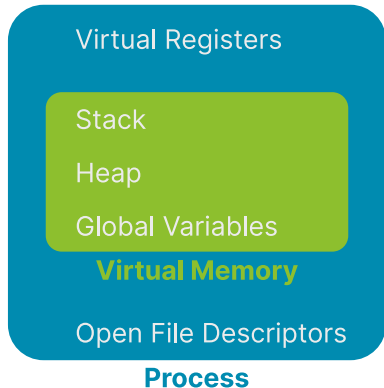
Running Processes On My Laptop



Recall: A Process is an Instance of a Running Program



We Can Add More to a Process



From The Kernel's Point of View

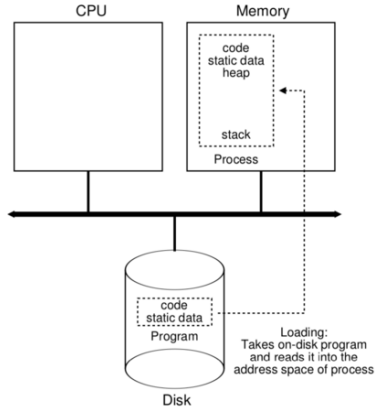


Figure 4.1: Loading: From Program To Process

A Process Control Block (PCB) Contains All Information

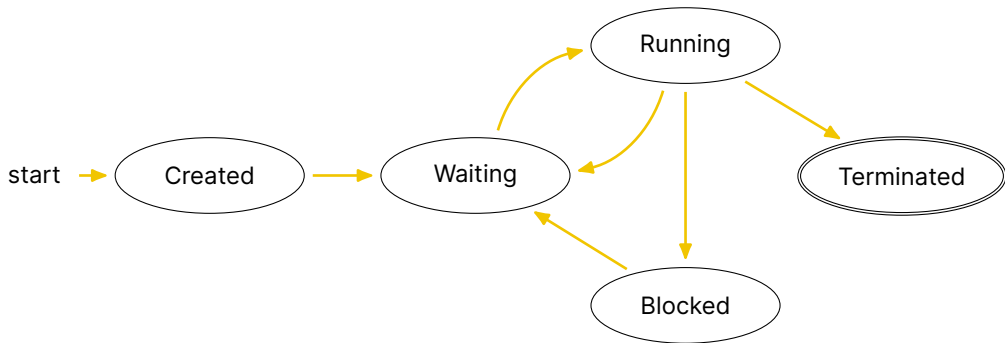
Specifically, in Linux, this is the [task_struct](#) you can browse on [GitHub](#)

It contains:

- Process state
- CPU registers
- Scheduling information
- Memory management information
- I/O status information
- Any other type of accounting information

Each process gets a unique process ID (pid) to keep track of it

Process State Diagram (You Could Rename Waiting to Ready)



You Can Read Process State Using the “proc” Filesystem

There's a standard /proc directory (on Linux) that represents the kernel's state
These aren't real files, they just look like it!

Every directory that's a number (process ID) in /proc represents a process

There's a file called status that contains the state (used for Lab 1)

We Could Create Processes from Scratch

We load the program into memory and create the process control block
(this is what Windows does)

Unix decomposes process creation into more flexible abstractions

Instead of Creating a New Process, We Could Clone It

Pause the currently running process, and copy it's PCB into a new one

This will reuse all of the information from the process, including variables!

Distinguish between the two processes with a parent and child relationship

They could both execute different parts of the program together

We could then allow either process to load a new program and setup a new PCB

fork Creates a New Process, A Copy of the Current One

`int` fork(`void`) as the following API:

- Returns the process ID of the newly created child process
 - 1: on failure
 - 0: in the child process
 - >0: in the parent process

There are now 2 processes running

Note: they can access the same variables, but they're separate
Operating system does "copy on write" to maximize sharing

fork Examples

```
#include <stdio.h>
#include <unistd.h>

int main() {
    int value = 2;
    fork();
    printf("Hello, world!\n");
    return 0;
}
```

fork Examples

```
#include <stdio.h>
#include <unistd.h>

int main() {
    int value = 2;
    if (fork() == 0)
        fork();
    fork();
    printf("Hello, world!\n");
    return 0;
}
```

fork Examples

```
#include <stdio.h>
#include <unistd.h>

int main() {
    int value = 2;
    int pid = getpid();
    fork();
    value = 3;
    printf("Hello, world, %d, %lx!\n", value, (long int)&value);
    return 0;
}
```

wait delays execution

wait as the following API:

- status: Address to store the wait status of the process
- Returns the process ID of child process
 - 1: on failure
 - >0: the child with a change

The wait status contains a bunch of information, including the exit code

Use `man wait` to find all the macros to query wait status

You can use `waitpid` to wait on a specific child process

wait Example

```
int main() {  
    int pid = getpid();  
    fork();  
    if (getpid() == pid){  
        // wait(NULL);  
        printf("Hello A\n");  
    }  
    else{  
        printf("Hello B\n");  
    }  
    return 0;  
}
```

execve Replaces the Process with Another Program, and Resets

execve has the following API:

- **pathname:** Full path of the program to load
- **argv:** Array of strings (array of characters), terminated by a null pointer
Represents arguments to the process
- **envp:** Same as argv
Represents the environment of the process
- Returns an error on failure, does not return if successful

execve-example.c Turns the Process into ls

```
int main(int argc, char *argv[]) {
    printf("I'm going to become another process\n");
    char *exec_argv[] = {"ls", NULL};
    char *exec_envp[] = {NULL};
    int exec_return = execve("/usr/bin/ls", exec_argv, exec_envp);
    if (exec_return == -1) {
        exec_return = errno;
        perror("execve failed");
        return exec_return;
    }
    printf("If execve worked, this will never print\n");
    return 0;
}
```

The Operating System Creates Processes

The operating system has to:

- Maintain process control blocks, including state
- Create new processes
- Load a program, and re-initialize a process with context